

SBC 201 Chapter 24 Glass and Glazing — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 4–12 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 24, source file Reference\SBC 201 2024\source_reference\Chapter_24 – GLASS AND GLAZING.txt.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **109** independently checkable numeric records (**88** Verified, **21** Verify source). Unresolved OCR (sloped-glass dead-load D for $\leq$ **30 degrees**, missing 2404.3.4, flattened Tables 2406.2(1)/(2), footnote attachment) is listed in the register and is not a design-release value.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, glazing specification, facade-engineering report, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from Chapter 16, Section 1605, Section 1607.9, Section 1609, Section 1609.2, SBC 301 Section 13.5.9, Chapter 26, Section 2607, Section 2610, Section 2110, Section 1015, Section 716, ASTM E1300, ASTM E2353, ASTM E2751, CPSC 16 CFR Part 1201 acceptance criteria beyond the category names stated here, ANSI Z97.1 class criteria beyond the class names stated here, AAMA/WDMA/CSA 101/I.S.2/A440 performance grades, commentary examples (including **6.4 mm** vs the code **6.5 mm**), bibliography years, or the existing chapter summary. Where Chapter 24 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Automatic sprinkler protection, mixed-use podium, storey count and construction type are unconfirmed. Chapter 24 does not branch on NFPA 13 versus 13R.
4. Unit-level versus common-area splits in 2405.3 Exceptions 4–5 and 2406.4.2 Exception 4 apply only inside individual Group R-2 dwelling units. They do not relax corridors, lobbies, amenity rooms or other public-use glazing.
5. The Group R-3 skylight-without-curb exception (2405.4) is not applied to this tower. Greenhouse screening Exception 3 and racquetball/squash court glazing (2408.2) are omitted from lead tables unless those uses are added.
6. Table 2406.2(1), Table 2406.2(2) and the sloped-glass dead-load formula for glass sloped **30 degrees or less** from horizontal are OCR-unresolved. No reconstructed category class or D-load is a design-release value. Section **2404.3.4** is missing from the extract (jump 2404.3.3 to 2404.3.5).
7. Glass thickness for wind and dead load is ASTM E1300 against Chapter 16; this chapter does not pick the millimetre for vertical or sloped lites except where a numbered thickness is stated (louvers, screens, guards, elevator vision panels).

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, load, sprinkler branch, mixed-use/podium condition or exception exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 24 points to another section/code/standard, or OCR/AHJ must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 24 source text or an unambiguous table cell.
Verify source	OCR, flattened table, page-split, missing subsection, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 24 application	Required project action
Facade / curtain-wall system	Unconfirmed: punched windows, stick/unitized curtain wall, point-fixed or fin-supported glass	2403.2 requires RDP analysis where an edge is not firmly supported; 2403.3 deflection limits apply only to firmly supported panes; 2404.1 sends thickness to ASTM E1300 / §1609	Freeze support condition (four-side vs two/three-side) on typical details before glass schedules
Construction type	Unconfirmed; high-rise R-2 is often Type I or II, not verified here	2405.4 requires noncombustible sloped-glazing/skylight frames in Types I and II	Lock construction type before specifying skylight sash material
Skylights / sloped glass	Unconfirmed amenity roofs, unit conservatories, canopy glass	2404.2 / 2405 apply only where slope is more than 15 degrees from vertical; screens, curbs and unit-skylight labels follow	Mark every lite > 15 degrees from vertical on plans; do not treat near-vertical curtain wall as 2405
In-unit vs common glazing	Units vs corridors, lobbies and amenity not scheduled	2405.3 Ex. 4–5 and 2406.4.2 Ex. 4 apply only inside individual R-2 dwelling units	Split the glass schedule into dwelling-unit and common-area packages
Glass guards vs in-fill	Unconfirmed balcony/stair/atrium guards	2406.4.4 makes all guard glass a hazardous location; 2407 adds laminated type, 6.5 mm , FoS four and top-rail rules	State laminated structural baluster vs post-and-in-fill on every glass-guard drawing
Wet rooms / amenity pool	Unit baths assumed; pool, spa, sauna, steam unconfirmed	2406.4.5 safety-glazes wet-facing glass with bottom < 1.5 m ; pool branch is Conditional	Tag shower/tub enclosures as safety glazing; freeze amenity wet-program before lobby/pool glass
Amenity gym	Unconfirmed	2408.3 Category II / Class A if a gym or basketball court exists; 2408.2 racquetball/squash is not opened here	Confirm whether a gym or court is in the program before applying 2408

Decision / gap	Current project basis	Why it controls Chapter 24 application	Required project action
Glass elevators / walkways	Unconfirmed	2409 laminated hoistway/car glass, 60 percent door glass, vision-panel 6 mm / 155–550 cm²	State whether hoistway doors, cars or floor glass are glazed; keep Chapter 16 / ASTM E2751 loads out of this matrix
Wind-borne debris region	Unconfirmed	2407.1.4 sends exterior glass rails to laminated Category II/Class A and §1609.2	Structural/wind consultant to confirm whether 1609.2 applies; do not import missile-test values here
Parking garage glass rails	Unconfirmed podium parking	2407.1.3 bans glass in garage handrails/guards except pedestrian areas not exposed to vehicles	Do not specify glass vehicle barriers in parking; keep any glass rail in a protected pedestrian-only zone
Sprinkler / mixed use / storeys	Unconfirmed	Chapter 24 has no occupancy, sprinkler or storey branch for these numbers	Do not wait on NFPA 13 vs 13R to apply 2403–2407; those decisions do not change this chapter
NOC / facade PE	Unconfirmed	ASTM E1300 thickness, SBC 301 §13.5.9 seismic drift and 1607.9 guard loads cannot be concluded from this extract	Engage the facade/structural engineer of record before freezing glass makeup

4. Scope, replacement and identification

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2401.1	Chapter glass and glazing scope	This chapter governs materials, design, construction and quality of glass, light-transmitting ceramic and light-transmitting plastic panels for exterior and interior vertical and sloped use. Plastic glazing shall also meet Chapter 26	All glass, ceramic and plastic glazing on the building	None stated	Direct	List glazing packages on the drawing index; send plastic skylights and plastic wall panels to Chapter 26 without importing 26 values here	Verified
2402.1	Replacement glass	Replacement glass shall be installed as required for new installations	Replacement, relocation or new glazing in existing openings	None stated	Direct	Specify replacement lites to the same 2403–2407 rules as new work; do not grandfather non-safety glass	Verified
2403.1	Pane type and thickness mark	Each pane shall bear the manufacturer's mark for type and thickness, unless the building official approves an affidavit that each light is glazed to approved documents. Tempered glass, except tempered spandrel, shall be permanently identified (acid etched, sand blasted, ceramic fired, laser etched, embossed, or equivalent). Tempered spandrel may use a removable paper mark. Safety glazing shall also be identified per 2406.3	Every lite on the building	Affidavit path is not available for tempered glass that must be permanently marked	Direct	Require visible type/thickness marks on shop drawings and mock-ups; permanent etch on tempered lites; paper mark only on tempered spandrel	Verified
2406.3	Safety-glazing designation	Each pane of safety glazing in a hazardous location shall carry a manufacturer's designation naming who applied it, the manufacturer or installer, the safety-glazing standard, and the 2403.1 information. The designation shall be permanent (acid etched, sand blasted, ceramic fired, laser etched, embossed, or equivalent). A complying label may replace the designation	Safety glazing in 2406.4 locations	Ex. 1: other than tempered glass, a certificate/affidavit may replace the designation if the building official approves. Ex. 2: tempered spandrel may use a removable paper designation	Direct	Show CPSC 16 CFR Part 1201 or ANSI Z97.1 marks on every hazardous-location lite; do not accept affidavits in lieu of tempered-glass marks	Verified
2406.3.1	Multipane assembly marking	Multipane glazed assemblies with individual panes not exceeding 0.01 m² exposed area shall have one or more panes marked per 2406.3; other panes shall be marked “CPSC 16 CFR Part 1201” or “ANSI Z97.1” as appropriate	Multipane decorative or leaded assemblies with lites ≤ 0.01 m²	Larger individual panes each need a full 2406.3 mark	Conditional	If decorative multipane doors or sidelights are used, mark at least one lite fully and stamp the remainder with the standard name	Verified
2406.1	Hazardous-location glazing	Individual glazed areas, including glass mirrors, in 2406.4 hazardous locations shall comply with 2406.1.1 through 2406.1.4	Glazing in doors, adjacent-to-door, large windows, guards, wet surfaces, stairs/ramps, or bottom landings	Mirrors and other glass panels mounted or hung on a surface that provides continuous backing support are not required to be safety glazed	Direct	Treat every 2406.4 location as safety glazing unless a numbered exception is demonstrated; backed bathroom mirrors may omit safety glass	Verified
2406.1.2–2406.1.4	Plastic, glass block, louvers	Plastic glazing shall meet ANSI Z97.1 weathering. Glass-block walls shall comply with Section 2110 . Louvered windows and jalousies shall comply with 2403.5	Plastic safety glazing, glass-block walls, or jalousie/louver windows	Louvers are not 2406 hazardous locations	External verification	Name ANSI Z97.1 weathering on plastic lites; send glass block to 2110; size louvers under 2403.5 — do not import 2110 or Chapter 26 numbers	Verified

5. Supports, framing, interior pinch and louvers

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2403.2	Unsupported-edge analysis	Where one or more sides of any pane are not firmly supported, or the pane is subjected to unusual load conditions, detailed construction documents, shop drawings and analysis or test data ensuring safe performance shall be prepared by a registered design professional	Two-side, three-side, point-fixed, fin-supported or otherwise free-edged glass	Four-edge firmly supported glass uses 2403.3 instead	Conditional	Send structural-glass and two-side-supported curtain-wall lites to the facade RDP; do not treat butt-glazed free edges as 2403.3 “firmly supported”	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2403.3	Firm-support framing deflection	To be firmly supported, framing for each pane shall limit deflection of the glass edge perpendicular to the pane to 1/175 of the glass edge length where that length is not more than 4050 mm , or 1/240 of the glass edge length + 6.25 mm where the edge length is greater than 4050 mm , under the larger of the positive or negative combined load from Section 1605	Four-edge supported windows, curtain wall, storefront and similar framed lites	Unsupported edges go to 2403.2	Direct	Put the 1/175 / 1/240 + 6.25 mm split on the curtain-wall specification; take combined loads from 1605 without copying 1605 values here	Verified
2403.4	Interior adjacent-pane pinch	Where interior glazing is adjacent to a walking surface, differential deflection of two adjacent unsupported edges shall be not greater than the thickness of the panels when 730 N/m is applied horizontally to one panel at any point up to 1.0 m above the walking surface	Interior butt-glazed or otherwise unsupported meeting edges next to a walkway	None stated	Direct	Check lobby, corridor and amenity interior glass joints for differential movement ≤ pane thickness under 730 N/m at ≤ 1.0 m AFF	Verified
2403.5	Louver and jalousie glass	Float, wired and patterned glass in louvered windows and jalousies shall be no thinner than nominal 5 mm and no longer than 1.2 m . Exposed glass edges shall be smooth. Wired glass with wire exposed on longitudinal edges shall not be used. Other glass types require building-official approval of the design	Louvered or jalousie windows	Louvers are not 2406 hazardous locations (2406.1.4)	Conditional	If louvers are specified, limit slat thickness to ≥ 5 mm and length to ≤ 1.2 m , with smooth edges and no exposed wire	Verified

6. Wind, seismic and dead-load design

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2404.1	Vertical glass wind design	Glass sloped 15 degrees or less from vertical in windows, curtain and window walls, doors and other exterior applications shall resist wind loads from basic design wind speed, V, in Section 1609 for components and cladding. Uniform-load resistance shall be determined in accordance with ASTM E1300 . Design shall satisfy Equation 24-1: 0.6 Fgw ≤ Fga	Exterior vertical and near-vertical glass (slope ≤ 15 degrees from vertical)	Thickness is not selected in this chapter	External verification	Size vertical lites to ASTM E1300 against 1609 C&C pressures; do not copy wind speeds or NFL charts into this matrix	Verified
2404.1	Glazed-wall seismic drift	Glass in glazed curtain walls, glazed storefronts and glazed partitions shall meet the seismic requirements of SBC 301, Section 13.5.9	Curtain wall, storefront and glazed partitions	Displacement limits are not stated in this chapter	External verification	Name SBC 301 §13.5.9 on the curtain-wall specification; do not import drift or clearance values from 301	Verified
2404.2	Sloped-glass load combinations	Glass sloped more than 15 degrees from vertical in skylights, sunrooms, sloped roofs and other exterior applications shall resist the most critical of Equations 24-2, 24-3 and 24-4: Fg = 0.6 Wo – D ; Fg = 0.6 Wi + D ; Fg = 0.3 Wi + D . Design shall satisfy Equation 24-5: Fg ≤ Fga , with F_ga from ASTM E1300 (short duration for 24-2 and 24-3; long duration for 24-4)	Exterior glass sloped more than 15 degrees from vertical	Unit skylights and tubular daylighting devices use 2405.5	External verification	Combine 1609 wind with glass dead load on sloped lites; do not import Wo / Wi values from Chapter 16	Verified
2404.2	Sloped-glass dead load D (≤30°)	The dead-load formula for glass sloped 30 degrees or less from horizontal is unreadable in this extract (LaTeX explosion after “D =”). No D value is adopted for that branch	Sloped glass with pitch ≤ 30 degrees from horizontal	The > 30-degree fragment is in the next row	External verification	Leave D blank on calculations until the published ≤ 30-degree formula is confirmed; do not reconstruct it from memory	Verify source
2404.2	Sloped-glass dead load D (>30°)	For glass sloped more than 30 degrees from horizontal, D = 0.0245 t_g cos θ (kN/m²), with θ the angle of slope from horizontal	Sloped glass with pitch more than 30 degrees from horizontal	≤ 30-degree D is unresolved	External verification	Use 0.0245 tg cos θ only for the > 30-degree branch after confirming tg units on the published page	Verified
2404.3.1	Vertical wired-glass load	Wired glass sloped 15 degrees or less from vertical shall satisfy Equation 24-6: 0.6 Fgw < 0.5 Fge , with F_ge the ASTM E1300 non-factored load for monolithic glass not thicker than the wired glass	Exterior vertical wired glass	Wired glass is outside ASTM E1300 scope; the 0.5 factor is the code adjustment	Conditional	If wired glass is used in vertical openings, apply Eq. 24-6; do not read NFL as F_ga without the 0.5 factor	Verified
2404.3.2	Sloped wired-glass load	Wired glass sloped more than 15 degrees from vertical shall resist 2404.2 combinations, with Fg < 0.5 Fge for Equations 24-2 and 24-3, and Fg < 0.3 Fge for Equation 24-4	Exterior sloped wired glass	None stated	Conditional	If wired glass is used in skylights or sloped roofs, apply Eqs. 24-7 and 24-8; do not use annealed NFL as the final resistance	Verified
2404.3.3	Vertical patterned-glass load	Patterned glass sloped 15 degrees or less from vertical shall satisfy Equation 24-9: Fgw < 1.0 Fge . F_ge shall be based on the thinnest part of the glass; interpolation between ASTM E1300 non-factored load charts is permitted	Exterior vertical patterned glass	Equation 24-9 is printed without the 0.6 factor used in 24-1, 24-6 and 24-12	Conditional	If patterned glass is used, take NFL from the thinnest section and apply Eq. 24-9 as printed — do not insert a 0.6 factor	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2404.3.4	Sloped patterned-glass load	No numbered 2404.3.4 paragraph is present (extract jumps from 2404.3.3 to 2404.3.5). Equations 24-10 and 24-11 are not adopted	Patterned glass sloped more than 15 degrees from vertical	None readable	External verification	Confirm the published sloped-patterned-glass equations before specifying patterned skylights; do not fill 24-10/24-11 from memory	Verify source
2404.3.5	Vertical sandblasted-glass load	Sandblasted glass sloped 15 degrees or less from vertical shall satisfy Equation 24-12: 0.6 F_{gw} < 0.5 F_{ge} . The F _{ge} value is for moderate levels of sandblasting	Exterior vertical sandblasted glass	None stated	Conditional	If sandblasted glass is used, apply Eq. 24-12 and confirm the “moderate sandblasting” NFL basis on the published standard	Verified
2404.4	Designs outside 2404	For designs outside the scope of this section, an analysis or test data for the specific installation shall be prepared by a registered design professional	Glass types or support conditions not covered by 2404.1–2404.3	None stated	Conditional	Route non-standard makeups (structural glass fins, cold-bent glass, unusual coatings) to the facade RDP	Verified

7. Sloped glazing and skylights

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2405.1	Sloped-glazing scope	Section 2405 applies to glass and other transparent, translucent or opaque glazing installed at a slope of more than 15 degrees from the vertical plane, including skylights, roofs and sloped walls	Any lite sloped more than 15 degrees from vertical	Vertical and ≤ 15-degree glass stays on 2404.1	Conditional	Tag every skylight, sloped roof glass and sloped atrium wall as 2405; keep near-vertical curtain wall on 2404.1	Verified
2405.2	Allowed sloped-glazing materials	Monolithic sloped glazing shall be laminated glass with a minimum 0.7 mm polyvinyl butyral (or equivalent) interlayer, wired glass, light-transmitting plastic meeting Section 2607 , heat-strengthened glass or fully tempered glass. Each layer of a multiple-layer system shall be one of those materials. Laminated glass and plastic in Items 1 and 2 shall not require 2405.3 screens or height restrictions. Plastic skylights: Section 2610 . Glass block: Section 2110.1	Sloped glazing and skylights	Annealed glass only under 2405.3 Exceptions 2 and 3	Conditional	Specify laminated (≥ 0.7 mm PVB), HS, FT, wired or Chapter 26 plastic; do not use annealed over occupied space unless Exception 2 or 3 is proven	Verified
2405.3	Broken-glass retention screen	Annealed, heat-strengthened, fully tempered and wired glass in monolithic sloped systems, and the same types used as the bottom layer over a walking surface in multiple-layer systems, shall have retention screens below the glass. Screens and fastenings shall support twice the weight of the glazing, be firmly fastened to the framing, and be installed within 100 mm of the glass. Material: noncombustible, not thinner than No. 12 B&S gage, mesh not larger than 25 mm by 25 mm . Corrosive atmospheres: structurally equivalent noncorrosive mesh	HS, FT, wired or annealed sloped glass over a walking surface	Laminated (2405.2) and plastic skip screens. Exceptions 1–5 below	Conditional	Show a No. 12 / 25 mm mesh within 100 mm of common-area sloped glass unless a numbered exception applies	Verified
2405.3 Ex. 1	Intervening-floor FT screen waiver	Fully tempered glass glazed between intervening floors at a slope of 30 degrees or less from vertical, with the highest point of the glass 3.0 m or less above the walking surface, may omit protective screens	FT sloped glass between floors, slope ≤ 30 degrees from vertical, top of glass ≤ 3.0 m	Does not waive screens on steeper or taller FT lites	Conditional	Use this waiver only on low, near-vertical FT between floors; keep screens on skylights above 3.0 m	Verified
2405.3 Ex. 2	Non-walking-surface screen waiver	Screens are not required below any glazing material, including annealed glass, where the walking surface below is permanently protected from falling glass or the area below is not a walking surface	Sloped glass over a protected or non-occupied void	Permanent protection must be demonstrated	Conditional	Omit screens only over inaccessible roofs, planted voids or documented catch systems — not over corridors or amenity floors	Verified
2405.3 Ex. 4	In-unit FT screen waiver	Screens shall not be required in individual dwelling units in Groups R-2, R-3 and R-4 where fully tempered glass is used as single glazing or as both panels of an IGU, and each pane is 1.5 m² or less, the highest point of the glass is 3.5 m or less above any walking surface or accessible area, and glass thickness is 5 mm or less	FT sloped glass inside an individual R-2 dwelling unit meeting all three limits	Does not apply to corridors, lobbies, amenity or other common areas	Direct	Allow in-unit FT skylights/sloped glass without screens only when area ≤ 1.5 m² , height ≤ 3.5 m and thickness ≤ 5 mm ; keep screens on common-area FT	Verified
2405.3 Ex. 5	In-unit laminated screen waiver	Screens shall not be required for laminated glass with a 0.4 mm polyvinyl butyral (or equivalent) interlayer in individual dwelling units in Groups R-2, R-3 and R-4 where each pane is 1.5 m² or less and the highest point of the glass is 3.5 m or less above a walking surface or other accessible area	Laminated sloped glass inside an individual R-2 dwelling unit meeting both limits	Does not apply to common areas. 2405.2 still uses 0.7 mm PVB as the general laminated minimum	Direct	Allow in-unit laminated sloped glass without screens only at ≤ 1.5 m² and ≤ 3.5 m ; do not use the 0.4 mm interlayer waiver in lobbies or amenity	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2405.4	Types I and II skylight frames	In Types I and II construction, sloped glazing and skylight frames shall be noncombustible. Where acid fumes deleterious to metal are incidental to the use, approved pressure-treated wood or other approved noncorrosive materials may be used. Framing shall resist tributary roof loads in Chapter 16	Sloped glazing and skylights on Type I or II buildings	Acid-fume noncorrosive branch	Direct	Specify noncombustible skylight frames on the tower; take frame loads from Chapter 16 without copying those loads here	Verified
2405.4	Skylight curb height	Skylights set at an angle of less than 45 degrees from the horizontal shall be mounted not less than 100 mm above the plane of the roof on a curb constructed as required for the frame. Skylights shall not be installed in the plane of the roof where the roof pitch is less than 45 degrees from the horizontal	Skylights on a roof flatter than 45 degrees	Group R-3 no-curb path is not applied to this R-2 tower. Unit skylights on roofs flatter than 14 degrees still need a 100 mm curb unless the manufacturer's instructions say otherwise	Conditional	Set skylights on a ≥ 100 mm curb on low-slope amenity and tower roofs; do not use the R-3 flush-to-roof exception	Verified
2405.5	Unit skylight and TDD label	Unit skylights and tubular daylighting devices shall be tested and labeled as complying with AAMA/WDMA/CSA 101/I.S.2/A440 . The label shall state manufacturer, approved labeling agency, product designation and performance grade. Separate positive/negative ratings shall comply with 2405.5.2; a single PG shall cover both pressures and comply with 2405.5.1 / Equation 24-13: $F_g \leq PG$	Factory-assembled unit skylights and TDDs	Site-built sloped glazing stays on 2404.2 / 2405.3	Conditional	Require the AAMA/WDMA/CSA label and a PG not less than F_g from 24-2 through 24-4; do not invent a PG value here	Verified
2405.5.2	Separate positive/negative PG	Skylights rated separately for positive and negative design pressure shall satisfy Equations 24-14 and 24-15: $F_{gi} \leq PGPo$ and $F_{go} \leq PGNe$. Branching uses 0.6 W_o compared with D to select which 2404.2 combinations apply	Unit skylights with separate +/- performance grades	Single-PG products use 2405.5.1	Conditional	Match labeled $PGPo$ / $PGNe$ to the 2404.2 inward and outward combinations; take W_o and D from 1609 / 2404.2 without importing those numbers	Verified

8. Safety-glazing test class

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2406.2	Default impact test	Where required by other sections, glazing shall be tested to CPSC 16 CFR Part 1201 and shall comply with Category II , unless otherwise indicated in Table 2406.2(1)	Hazardous-location glazing under 2406.4	Glazing not in doors or hot-tub / whirlpool / sauna / steam / bathtub / shower enclosures may be tested to ANSI Z97.1 and shall comply with Class A , unless otherwise indicated in Table 2406.2(2)	Direct	Specify CPSC Category II as the default; allow ANSI Class A only off the door/wet-enclosure list. Do not read Table 2406.2 category letters from the flattened extract	Verified
Table 2406.2(1)	CPSC category by area	Minimum CPSC 16 CFR Part 1201 category classification by exposed surface area and location. Area rows 0.84 m² or less and more than 0.84 m² are readable. Category-class cells are flattened concatenated HTML and are not adopted	Locations listed in the table (storm/combination doors, doors, 2406.4.3 panels, 2406.4.2 panels, 2406.4.5 enclosures, sliding patio doors)	Default remains Category II until the published table is confirmed	External verification	Confirm the published Table 2406.2(1) before accepting Category I on any lite; do not parse I INo requirement IIII from this extract	Verify source
Table 2406.2(2)	ANSI class by area	Minimum ANSI Z97.1 class by exposed surface area and location. Area rows 0.84 m² or less and more than 0.84 m² are readable. Class cells are flattened and are not adopted. Use is only permitted by the exception to 2406.2	2406.4.3 panels, 2406.4.2 panels, and 2406.4.5 doors/enclosures on the ANSI path	Default remains Class A until the published table is confirmed	External verification	Confirm the published Table 2406.2(2) before accepting Class B; do not adopt concatenated A/B tokens	Verify source

9. Hazardous locations

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2406.4.1	Door glazing	Glazing in all fixed and operable panels of swinging, sliding and bifold doors is a hazardous location	Any glazed swinging, sliding or bifold door	Ex. 1: opening through which a 75 mm sphere cannot pass. Ex. 2: decorative glazing. Ex. 3: curved glazed panels in revolving doors. Ex. 4: commercial refrigerated-cabinet doors	Direct	Specify safety glazing in unit entry doors, balcony doors and common-area glazed doors unless the lite is smaller than a 75 mm sphere or a named exception applies	Verified
2406.4.2	Glazing adjacent to doors	Glazing in an individual fixed or operable panel adjacent to a door is a hazardous location where the nearest vertical edge is within a 600 mm arc of either vertical door edge in the closed position and the bottom exposed edge is less than 1.5 m above the walking surface	Sidelights and adjacent panels next to doors	Ex. 1: decorative glazing. Ex. 2: intervening wall or other permanent barrier. Ex. 3: closet/storage 900 mm or less in depth (then 2406.4.3). Ex. 4: latch-side walls perpendicular to the closed door in one- and two-family dwellings or within dwelling units in Group R-2 — not in public-use areas	Direct	Safety-glaze corridor and lobby sidelights inside the 600 mm / 1.5 m envelope. Inside dwelling units only, latch-side perpendicular walls may omit 2406.4.2	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2406.4.3	Large low window lites	Glazing in an individual fixed or operable panel is a hazardous location when all of the following are met: exposed area of an individual pane greater than 0.85 m² ; bottom edge less than 450 mm above the floor; top edge greater than 900 mm above the floor; one or more walking surfaces within 900 mm horizontally of the plane of the glazing	Full-height or low-sill glazed panels next to a walkway	Ex. 1: decorative glazing. Ex. 2: horizontal rail on the accessible side(s) 850 mm to 950 mm AFF, capable of 730 N/m without contacting the glass, not less than 40 mm in cross-sectional height. Ex. 3: outboard IGU (or multiple-glazing) pane where the bottom exposed edge is 7.5 m or more above grade, roof, walking surface or other horizontal or sloped (within 45 degrees of horizontal) surface adjacent to the glass exterior	Direct	Safety-glaze unit and common full-height glass that meets all four triggers, or provide the 850–950 mm / 40 mm rail. Outboard IGU panes above 7.5 m may omit safety glass; inboard panes still follow 2406.4.3	Verified
2406.4.4	Guard and railing glass	Glazing in guards and railings, including structural baluster panels and nonstructural in-fill panels, is a hazardous location regardless of area or height above a walking surface	Any glass in a guard or railing	None stated. Material rules are in 2407	Direct	Specify safety glazing for every balcony, stair and atrium glass guard, including in-fill — no area or height waiver	Verified
2406.4.5	Wet-surface glazing	Glazing in walls, enclosures or fences containing or facing hot tubs, spas, whirlpools, saunas, steam rooms, bathtubs, showers and indoor or outdoor swimming pools, where the bottom exposed edge is less than 1.5 m above any standing or walking surface, is a hazardous location. Applies to single glazing and all panes in multiple glazing	Unit baths/showers (Direct); amenity pool/spa/sauna/steam (Conditional)	Glazing more than 1.5 m horizontally in a straight line from the water’s edge of a bathtub, hot tub, spa, whirlpool or swimming pool	Direct	Safety-glaze shower and tub enclosures and facing walls to 1.5 m AFF, all IGU panes. Distance > 1.5 m from the water’s edge may drop this section (other 2406.4 rules still apply)	Verified
2406.4.6	Stair and ramp glazing	Glazing where the bottom exposed edge is less than 1.5 m above the plane of the adjacent walking surface of stairways, landings between flights and ramps is a hazardous location	Glass beside stairs, intermediate landings and ramps	Ex. 1: side with a guard complying with Sections 1015 and 1607.9 , and the plane of the glass greater than 450 mm from the railing. Ex. 2: glazing 900 mm or more horizontally from the walking surface	Direct	Safety-glaze stair-adjacent glass below 1.5 m , or keep glass > 450 mm behind a 1015/1607.9 guard, or > 900 mm horizontally from the walking surface. Do not import 1015/1607.9 geometry	Verified
2406.4.7	Bottom-landing glazing	Glazing adjacent to the landing at the bottom of a stairway is a hazardous location where the glazing is less than 1.5 m above the landing and within a 1.5 m horizontal arc that is less than 180 degrees from the bottom-tread nosing	Glass at the bottom stair landing	Guard complying with 1015 and 1607.9 where the plane of the glass is greater than 450 mm from the guard	Direct	Safety-glaze glass in the 1.5 m / < 180-degree arc off the bottom nosing, or set it > 450 mm behind a complying guard	Verified
2406.5	Fire-department access panels	Fire department glass access panels shall be of tempered glass. For insulating glass units, all panels shall be tempered glass	FD glass access panels in the envelope	None stated	Conditional	If FD access panels are glazed, specify tempered glass in every pane of the IGU	Verified

10. Glass in handrails and guards

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2407.1	Guard and handrail glass type	Glass used in a handrail or guard shall be laminated glass of fully tempered or heat-strengthened glass, complying with Category II of CPSC 16 CFR Part 1201 or Class A of ANSI Z97.1, and shall be an approved safety glazing material per 2406.1.1. Minimum nominal thickness for all glazing types: 6.5 mm	Glass balusters, in-fill and glass handrails	Single fully tempered Category II / Class A is permitted where there is no walking surface beneath or the walking surface is permanently protected from falling glass	Direct	Specify laminated FT or HS, Category II/Class A, nominal ≥ 6.5 mm on balcony and stair glass guards. Use the code 6.5 mm , not commentary 6.4 mm . Monolithic FT only over a protected or non-occupied void	Verified
2407.1.1	Guard glass design load	Glass handrails and guards and their support systems shall be designed to withstand the loads in Section 1607.9 . Glass handrails and guards shall be designed using a factor of safety of four	Glass guards and handrails	1607.9 load values are not in this chapter	External verification	Require FoS four on glass-guard calculations; take live loads from 1607.9 without copying those loads here	Verified
2407.1.2	Structural glass balusters	Guards with structural glass balusters (posts, columns or panels) shall have an attached top rail or handrail supported by not fewer than three glass balusters, or otherwise supported to remain in place should one baluster fail	Structural glass baluster guards	An attached top rail/handrail is not required where laminated glass has two or more plies of equal thickness and the same glass type, and the balusters are tested to remain in place as a barrier after impact or breakage per ASTM E2353	Direct	Provide a top rail on ≥ three balusters, or use equal-ply laminated glass with an ASTM E2353 post-breakage report	Verified
2407.1.3	Parking-garage glass rails	Glazing materials shall not be installed in handrails or guards in parking garages except for pedestrian areas not exposed to impact from vehicles	Parking-garage handrails and guards	Pedestrian-only areas not exposed to vehicles	Conditional	Ban glass vehicle barriers in podium parking; limit any glass rail to a protected pedestrian walk	Verified
2407.1.4	Wind-borne-debris glass rails	Glazing in exterior handrails or guards in wind-borne debris regions shall be laminated glass complying with Category II of CPSC 16 CFR Part 1201 or Class A of ANSI Z97.1. Where the top rail is supported by glass, the assembly shall be tested to the impact requirements of Section 1609.2 and the top rail shall remain in place after impact	Exterior glass rails in a wind-borne debris region	1609.2 missile criteria are not stated in this chapter	Conditional	If 1609.2 applies, specify laminated Category II/Class A exterior rails and a 1609.2 assembly test for top-rail-on-glass — do not import missile speeds here	Verified

11. Walkways and elevator glass

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2409.1	Glass walkways	Glass installed as part of a floor/ceiling assembly as a walking surface and constructed with laminated glass shall comply with ASTM E2751 or with the load requirements in Chapter 16 . Assemblies shall comply with the fire-resistance rating and marking requirements of this code where applicable	Laminated glass floors, landings or ceiling walkways	Non-laminated glass walkways need an engineered Chapter 16 design; those loads are not imported	Conditional	If a glass floor or skywalk is proposed, name ASTM E2751 or Chapter 16 on the structural sheet; do not pick kN/m² from this chapter	Verified
2409.2	Hoistway enclosure glass	Glass in elevator hoistway enclosures and hoistway doors shall be laminated glass conforming to ANSI Z97.1 or CPSC 16 CFR Part 1201	Glass hoistway walls or doors	Fire-resistance-rated hoistways also comply with Section 716 (2409.2.1)	Conditional	Specify laminated safety glass on glass hoistways; coordinate rated hoistways with 716 without importing 716 minutes	Verified
2409.2.2	Glass hoistway-door area	Glass in glass hoistway doors shall be not less than 60 percent of the total visible door panel surface area as seen from the landing side	Glass hoistway doors	None stated	Conditional	Keep visible glass ≥ 60 percent of the landing-side door panel on glass hoistway doors	Verified
2409.3	Hoistway-door vision panels	Glass in vision panels in elevator hoistway doors may be any transparent glazing not less than 6 mm thick conforming to ANSI Z97.1 Class A or CPSC 16 CFR Part 1201 Category II. Area of any single vision panel shall be not less than 155 cm² . Total area of one or more vision panels in any hoistway door shall be not more than 550 cm²	Vision panels in (otherwise opaque) hoistway doors	None stated	Conditional	Size each vision lite ≥ 6 mm , ≥ 155 cm² and keep the door total ≤ 550 cm² ; specify Class A / Category II	Verified
2409.4.1	Elevator-car glass type	Glass in elevator car enclosures, glass elevator car doors and glass lining walls and ceilings of elevator cars shall be laminated glass conforming to ANSI Z97.1 Class A or CPSC 16 CFR Part 1201 Category II	Glass elevator cars	Tempered glass may line walls and ceilings if (1) bonded to a non-polymeric coating, sheeting or film that holds fragments, (2) not further treated (sandblasting, etching, heat treatment or painting) in a way that alters original properties, and (3) tested to Class A / Category II laminated-glass acceptance criteria	Conditional	Specify laminated Class A / Category II for glass cars; allow filmed tempered linings only with the three numbered conditions	Verified
2409.4.2	Glass elevator-car door area	Glass in glass elevator car doors shall be not less than 60 percent of the total visible door panel surface area as seen from the car side	Glass elevator car doors	None stated	Conditional	Keep visible glass ≥ 60 percent of the car-side door panel	Verified

12. Athletic-facility glazing

Amenity gym/basketball is opened in the gap register as unconfirmed. Racquetball/squash (2408.2) is not opened and is omitted from this table.

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2408.1	Athletic-facility glazing scope	Glazing in athletic facilities and similar uses subject to impact loads, forming whole or partial wall sections or used as a door or part of a door, shall comply with Section 2408	Gym, court or similar impact-use rooms	2408.2 racquetball/squash is not applied unless dedicated courts are added	Conditional	If an amenity gym is programmed, route wall and door glass through 2408.3	Verified
2408.3	Gym and basketball glass	Glazing in multipurpose gymnasiums, basketball courts and similar athletic facilities subject to human impact loads shall comply with Category II of CPSC 16 CFR Part 1201 or Class A of ANSI Z97.1	Gym, basketball court or similar amenity	None stated	Conditional	Specify Category II / Class A on gym wall and door glass; do not use Category I / Class B in those rooms	Verified

13. Project-use controls

1. Use **Verified** rows for coordination after the row trigger and unit-versus-common branch are confirmed.
2. Treat every **Verify source** row (2404.2 D for slope ≤ **30 degrees**, missing 2404.3.4, Tables 2406.2(1) and 2406.2(2), and footnote attachment after those tables) as a design hold point. No affected value is to be placed on issued-for-approval drawings without a published-source check.
3. Do not import ASTM E1300 non-factored loads, Chapter 16 / 1605 / 1607.9 / 1609 pressures, SBC 301 §13.5.9 drift limits, Chapter 26 plastic criteria, Section 2110 glass-block limits, Section 1015 guard geometry, or AAMA/WDMA/CSA performance grades from this matrix.
4. Do not apply 2405.3 Exceptions 4–5 or 2406.4.2 Exception 4 to corridors, lobbies, amenity or other public-use glazing.
5. Do not apply the Group R-3 skylight-without-curb exception, greenhouse Exception 3, or 2408.2 racquetball/squash numbers to this tower unless those uses are added to the gap register.
6. Do not adopt commentary **6.4 mm** in place of the code **6.5 mm** guard-glass thickness, and do not reconstruct Table 2406.2 Category I/II or Class A/B cells from the flattened HTML.
7. Record facade support condition, skylight presence, glass-guard type and amenity wet/gym decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

14. Coverage summary

Internal inventory of the attached Chapter 24 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)

- **Total independently checkable numeric records:** 109
- **Verified:** 88
- **Verify source:** 21

Counts by top-level section

Top-level section	Records
2401	0
2402	0
2403	7
2404	21
2405	21
2406	47
2407	3
2408	5
2409	5

Appended-table coverage

Appended table	Records	Verify source records
Table 2406.2(1)	14	12
Table 2406.2(2)	8	6
Footnotes after Table 2406.2(2) (attachment unresolved)	1	1

Table area-row tokens (**0.84 m² or less / more than 0.84 m²**) are Verified (2 per table). Category/class cells are Verify source.

Coverage cross-check against SBC 201 Chapter 24 Glass and Glazing (2024)_CS.md was topics-only: 2403 framing/interior deflection; 2404 ASTM E1300 / 15° split; 2405 screens and R-2 dwelling-unit exceptions; 2406.4 hazardous-location list; 2407 laminated **6.5 mm** guards; 2409 elevator glass. No CS.md value was copied into a matrix cell.

General Comments, Purpose, bibliography years, Commentary Figures 2403.1 / 2405.3 / 2406.4, and the mis-bound Chapter 23 AITC certificate on page 1783 were not inventoried.

15. Unresolved-source register

Hold points for the 21 **Verify source** inventory records. Counts are record counts, not distinct numeric values. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
2404.2 dead load D, slope ≤ 30 degrees from horizontal	1 record. Formula after “D =” is destroyed LaTeX (t_g repeated). Only the > 30-degree fragment 0.0245 t_g cos θ is readable	Do not reconstruct D for the ≤ 30-degree branch from IBC memory; confirm the published formula before sizing low-pitch sloped glass
2404.3.4 sloped patterned glass	1 record. Numbering jumps 2404.3.3 → 2404.3.5. Equations 24-10 and 24-11 are absent	Confirm published sloped-patterned-glass equations before specifying patterned skylights
Table 2406.2(1)	12 flattened CPSC category cells (storm/combination doors, doors, 2406.4.3, 2406.4.2, 2406.4.5, sliding patio × two area rows). Concatenated string mixes I , II and No requirement	Default remains Category II (2406.2). Do not parse category letters from the HTML blob
Table 2406.2(2)	6 flattened ANSI class cells (2406.4.3, 2406.4.2, 2406.4.5 × two area rows)	Default remains Class A on the 2406.2 exception path. Do not adopt concatenated A/B tokens
Footnotes a–c after Table 2406.2(2)	1 numeric footnote (0.84 m² in note a) appears to belong to Figure 2403.1 identifying-mark notes (I/II after 16 CFR 1201; SGCC U/L), not to the classification tables	Do not treat figure-mark notes as Table 2406.2 classification footnotes